DEVSECOPS LAB BOOK · PRACTICAL DEPLOYMENT

Chapter 34

Implementation & Deployment Guide

From a 30-minute rapid deploy to a full 13-phase build-out

A practical walkthrough of deployment phases, customization, troubleshooting, and recovery — for instructing teams standing up the lab environment.

SESSION OVERVIEW

What We'll Cover



Rapid Deployment

The 5-command path from clone to a running cluster in 30 minutes.



Deployment Roadmap

Six phases, from network foundation through advanced container patterns.



Phase Details

Time budget, VM counts, and validation criteria for every phase.



Total Effort

Where the hours go across the full build-out.



Customization

Adapting network, domain, scale, and resources to your environment.



Troubleshooting

Fast diagnostic commands for the most common failure points.



Recovery

Backing up VMs, etcd, and Vault — and proving restores actually work.

RAPID DEPLOYMENT

Zero to Cluster in 30 Minutes

Five commands take you from a fresh clone to a validated Kubernetes cluster.

1 Clone Repository

```
git clone .../lab
~/lab
cd ~/lab
```

2 Init Terraform

```
cd opentofu
terraform init
```

3 Deploy Infra

```
terraform apply
-auto-approve
```

4 Configure

```
cd ../ansible
ansible-playbook
site.yml
```

5 Validate

```
kubectl get nodes
```

Prerequisite: libvirt/KVM hypervisor, Terraform (OpenTofu), and Ansible installed on the deployment host.

DEPLOYMENT ROADMAP

Six Phases, One Lab Environment

Phase 1	**Foundation**	DNS, Identity, HTTPS — 7 VMs. All certificates and networking must work first.	8–10 hrs
Phase 2	**Automation (IaC)**	Packer, Terraform, Ansible. Deliverable: 30-minute full lab deployment.	6–8 hrs
Phase 3	**Security**	Vault + Teleport integration. All access must be auditable.	8–10 hrs
Phase 4	**CI/CD (Supply Chain)**	GitLab, scanning, compliance. The most complex phase.	12–14 hrs
Phase 5	**Advanced (HA, DR)**	High-availability etcd, clustering, disaster recovery procedures.	8–10 hrs
Phase 6–13	**Container & Advanced**	Kubernetes, Helm, Istio, GitOps, monitoring, advanced patterns.	40–50 hrs

PHASES 1–3

Foundation, Automation, Security

8–10 hrs

Phase 1 — Foundation

VMS

7 (DNS, Identity, LB, Proxy, Auth, Vault, VCS)

KEY REQUIREMENT

All certificates and networking must work first

6–8 hrs

Phase 2 — Automation (IaC)

TOOLS

Packer, Terraform, Ansible

DELIVERABLE

30-minute full lab deployment

8–10 hrs

Phase 3 — Security

FOCUS

Vault + Teleport integration

VALIDATION

All access must be auditable

PHASES 4–13

CI/CD, Advanced Ops, Container Platform

12–14 hrs

Phase 4 — CI/CD (Supply Chain)

INCLUDES

GitLab, scanning, compliance

NOTE

The most complex phase in the build-out

8–10 hrs

Phase 5 — Advanced (HA, DR)

FOCUS

High-availability etcd, clustering

INCLUDES

Disaster recovery procedures

40–50 hrs

Phase 6–13 — Container & Advanced

COVERS

Kubernetes, Helm, Istio, GitOps, monitoring

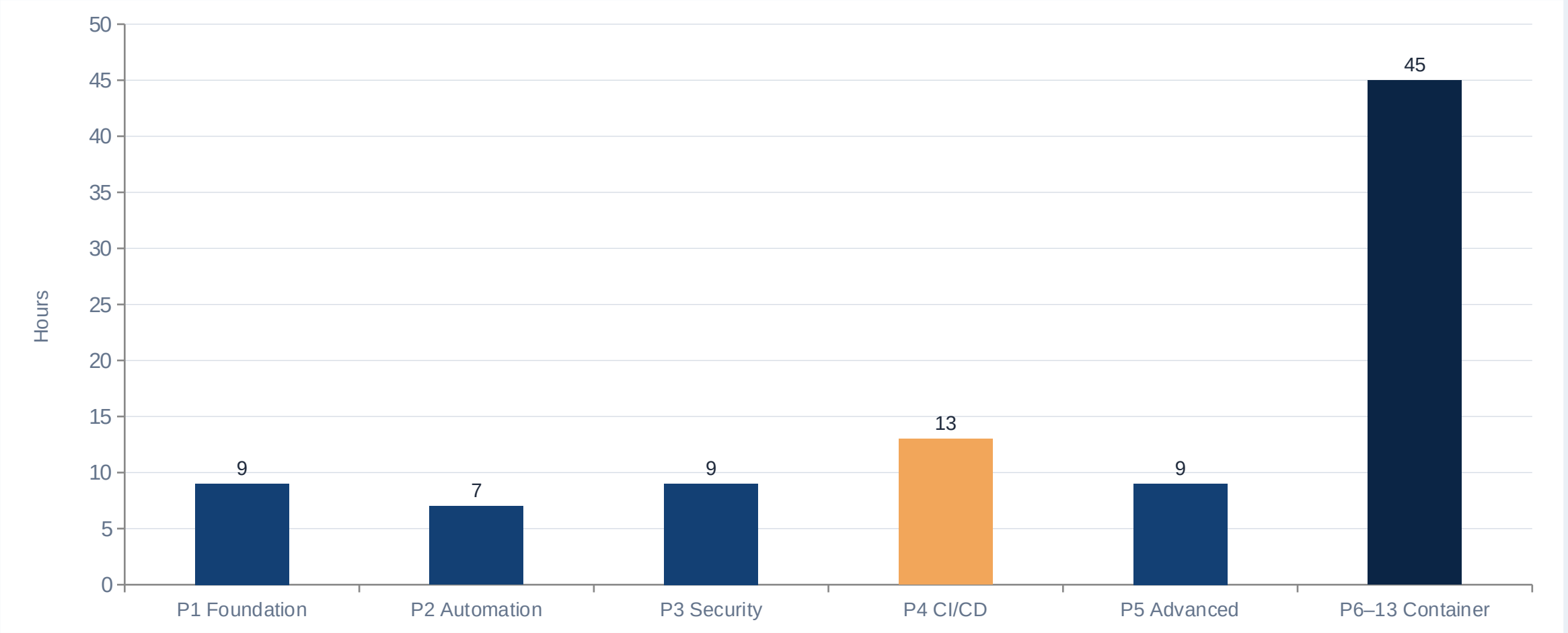
INCLUDES

Advanced patterns and compliance

EFFORT PLANNING

Where the Hours Go

Midpoint hour estimates by phase — useful for staffing a build-out sprint.



Total build-out: roughly 82–102 hours end to end — Phase 4 (CI/CD) and Phases 6–13 (container platform) dominate the schedule.

CUSTOMIZATION

Adapting the Lab to Your Environment



Change Network

```
sed -i
's/10.0.2.0\/24/YOUR_CIDR/g'
*.tf
```



Change Domain

```
sed -i
's/devsecops.com/YOUR_DOMAIN/
g' *.yaml
```



Scale Workers

```
# Edit worker_count in
# terraform.tfvars
```



Adjust Resources

```
# Modify CPU/RAM/disk
# in IaC variables
```

TROUBLESHOOTING

Fast Diagnostic Commands

Symptom	Command	What It Tells You
VMs not starting	`virsh list --all`	Confirms VM state and hypervisor visibility
Network issues	`virsh net-list`	Lists active libvirt networks
DNS problems	`nslookup devsecops.com 10.0.2.10`	Queries the lab DNS server directly
K8s not ready	`kubectl describe nodes`	Surfaces node conditions and scheduling events
Terraform errors	`terraform state list | terraform show`	Inspects current state before re-applying

RECOVERY

Backup Before You Need It

Backup VMs

```
virsh dumpxml cp-01 > \
/mnt/backups/cp-01.xml
```

Backup etcd

```
ETCDCTL_API=3 etcdctl \
snapshot save /mnt/backups/etcd.db
```

Backup Vault

```
vault write \
sys/storage/raft/snapshot \
file=@vault-snapshot
```

Test recovery regularly. A backup you haven't restored from is a guess, not a guarantee.

WRAP-UP

From Clone to Cluster to Confidence

Keep in your runbook:

- The 5-command rapid deploy for fresh environments
- Phase order — foundation and networking always come first
- sed one-liners for network, domain, and scale changes
- The five troubleshooting commands, memorized
- A recurring, tested backup and recovery schedule

Fastest Path to Green

git clone → terraform init → terraform apply → ansible-playbook → kubectl get nodes.

30 minutes, five commands, one validated cluster.

Discussion: which phase would you compress first if the timeline got cut in half?